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Impact of entrepreneurial leadership on project success: mediating role of knowledge management processes

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Abstract

Purpose – Drawing on the knowledge-based view (KBV), the study investigates the impact of entrepreneurial leadership (EL) on knowledge management (KM) processes and further examines the mediating role of KM processes on the linkage between EL and project success (PS).
Design/methodology/approach – Survey data were collected from 304 project workers in software projects, and the proposed relationships were assessed through SMART-PLS structural equation modeling tool.
Findings – The study found a significant impact of EL on KM processes and PS. The analysis also revealed that KM processes significantly impact project success while EL impact PS indirectly through KM processes.
Originality/value – The relevancy of the research stems from the scarcity of research on EL, while studies on the role of leadership as a predictor of KM are significantly limited. Additionally, there is a scarcity of research on the impact of KM on project success. This is one of the earliest studies that investigate the inter-relationship among EL, KM processes and project success.

Keywords Entrepreneurial leadership, Knowledge management Processes, Project success
Paper type Research paper

Introduction

Business practices around the world have raised some difficult issues for managers with respect to leadership styles. Leaders should concentrate on taking initiatives, risk-taking and building competitiveness in order to achieve goals and guarantee a sustained competitive advantage (Gil *et al.*, 2018; Villaluz and Hechanova, 2019). The knowledge-based view (KBV) (Grant, 1996) recognizes knowledge as a critical asset which, if managed effectively, can lead an organization towards the achievement of competitive advantage and better project performance. Leaders are the core assets to manage knowledge effectively (Bryant, 2003). Leaders have an uninterrupted impact on how the organizations should approach and manage KM processes (Singh, 2008). KBV assumes that KM processes such as knowledge acquisition, creation, storage, sharing/transform and knowledge application/utilization play a critical role in achieving better performance and competitive advantage (Shujahat *et al.*, 2019). 21st-century organizations keep on confronting changes particularly with the improvement and speeding up of innovation, ordinary leadership may not be an alternative and EL is the response to survival and accomplishment for present-day organizations (Abubakar *et al.*, 2018).

EL is an emerging paradigm and is the blend of business entrepreneurship and leadership which at whatever point associated will enable the organization to exploit the opportunity to



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attain the upper hand (Huang *et al.*, 2014; Abubakar *et al.*, 2018). EL as the leadership style and behavior can encourage development and performance in unverifiable, challenging and aggressive conditions (Al Mamun *et al.*, 2018; Strobl *et al.*, 2018). The need for entrepreneurial leaders in businesses is increasing and the subject of EL requires further attention of researchers (Al Mamun *et al.*, 2018).

The need to study entrepreneurial leadership stems from the fact that scholars have shown concern towards the effectiveness of general leadership styles (e.g. transformational/transactional) in directing employees' involvement in the challenging tasks of project management for two main reasons. First, general leadership styles are not specifically designed to regulate employees' behavior and second, they do not determine the leader's exact behavior and techniques to direct different complex project-related processes (Andersen *et al.*, 2006). Leaders of high technology businesses, defined as the businesses that allocate significantly high financial resources to scientific, technological, research and development activities (Butchart, 1987), need to develop new leadership competencies in order to effectively manage the processes in their business (Chen *et al.*, 2014). An increasing number of studies have highlighted entrepreneurial leadership as the leadership style and behavior that significantly stimulates and fosters innovation and opportunity recognition in highly challenging, turbulent and competitive environments (Freeman and Siegfried, 2015). Yet, there is a considerable lack of research on the effects of this leadership style in project-based organizations and the success of the projects.

Organizational leadership has been identified as a prominent KM enabler (Iqbal *et al.*, 2019; Obeidat *et al.*, 2016). The emerging EL style has gained particular interest in performance management research in project-based organizations due to their relevance to the project-based organizations and their management (Abubakar *et al.*, 2018; Mishra and Misra, 2017; Zainol *et al.*, 2018). Scholars have described and assessed KM enablers and KM processes as KM capability (Iqbal *et al.*, 2019;Al-Hakim and Hassan, 2011). KM processes include practices pertinent to knowledge acquisition, creation, sharing and utilization that foster sustainable competitive advantage (Torres *et al.*, 2018). Existing literature has identified a number of gaps in relation to the role of EL, KM processes and project success that shall be addressed.

First, although the interest in EL has been growing, the concept of EL is still significantly under-developed (Dean and Ford, 2017; Harrison *et al.*, 2018; Strobl *et al.*, 2018). The various parts of leadership literature have yet to permeate the field of entrepreneurship (Leitch and Volery, 2017).

Second, researchers have recently begun to investigate the role of leaders in KM (Matošková *et al.*, 2018; Naqshbandi and Jasimuddin, 2018; Donate and de Pablo, 2015). Existing research has assessed the impact of leadership on a limited number of KM processes (Bavik *et al.*, 2018). Masa'deh *et al.* (2017) stressed that the KM processes, in particular, knowledge sharing, has been well researched. Hence, the present study has taken into consideration four KM processes: Knowledge Acquisition, Knowledge Storage, Knowledge Sharing, and Knowledge Utilization.

Third, the relationship between leadership and organizational performance is studied in light of different mediating variables (Fu *et al.*, 2018; Kantur, 2016; Al-Dhaafri *et al.*, 2016). The direct relationship between EL and project success can be questionable. Existing research has called for an assessment of KM processes as a mediating variable that could explain the impact of leadership on different organizational level outcomes (Chang *et al.*, 2018; Nasab and Afshari, 2019; Naqshbandi and Jasimuddin, 2018).

Fourth, the impact of KM has been investigated on project-based outcomes (Project Performance and Success). Existing research has evaluated the impact of a limited number of KM processes on project success (see Yang *et al.*, 2012; Fedor *et al.*, 2003). Showing that

research on knowledge management in project management literature is still in its infancy (Bavik *et al.*, 2018).

This research intends to fill the identified knowledge gaps and in doing so makes several contributions. First, the study extends the limited research on the understanding of EL (Cai, 2018) and its impact on project success. Our study is among the first to consider EL as an important antecedent of project success. Second, the study investigates whether EL can serve as an important predictor of KM processes as the role of leadership as a predictor of KM processes is significantly limited (Bavik *et al.*, 2018; Shamim *et al.*, 2019). Third, the study further assesses whether KM processes mediate the relationship between EL and project success. This would help in providing a basis for understanding the mediating mechanism through which EL affects knowledge management processes and ultimately result in a successful project. Understanding the mechanisms that lead to the effect of EL on project success would help to articulate a better theoretical understanding of this relationship. Fourth, there is significantly limited research on how different KM processes impact project success. The research would enrich the knowledge management literature and help understand the role of KM in fostering improved project success. The research contributes to the leadership literature by assessing a new form of leadership style and its implication towards knowledge management and project success. Additionally, the study would provide further insights to KBV by demonstrating how EL contributes towards KM processes that lead to a successful project. Through the integration of KBV, we identify the KM process as a distinct pathway through which EL is associated with project success.

Literature review

Entrepreneurial leadership (EL)

Cunningham and Lischeron (1991) began the research on EL who suggested that it refers to setting clear goals, creating chances, allowing people, safeguarding organizational understanding and creating human resource frameworks. Recently, Renko *et al.* (2015) stressed that EL involves inducing and directing the performance of group members towards the attainment of organizational goals through recognition and exploitation of entrepreneurial opportunities. Chen (2007) depicts that EL helps to enhance the success of the leader's task achievement and efficiency. Conceptualized by McGrath and Macmillan (2000) who stressed that the competitive pressures and increasing uncertainty in the globalized dynamic market require a new type of leader. Thus, they gave the description of the "entrepreneurial leader" as the one who has an apt "entrepreneurial" approach and precisely the ability to keep abreast with the fast-changing markets, and to exploit opportunities to gather advantages before and faster than others (Karmarkar *et al.*, 2014). Entrepreneurial leadership is a unique leadership style that is focused on making heterogeneous talents work more creatively and innovatively in collective processes to respond to the uncertain business environment to create a coherent strategy and novel outcomes. Hence, organizations need to be able to measure and thus manage their entrepreneurial leadership (Musa and Fontana, 2016).

Entrepreneurs operating enterprises function as the leader of their business and need specific leadership skills to launch, grow, and sustain a successful venture. In the increasingly turbulent and competitive environment business firms face today, a type of "entrepreneurial" leader distinct from other behavioral forms of leadership is required (Gupta *et al.*, 2004). Transformational leaders demonstrate some features and behaviors that also characterize entrepreneurial leaders; hence, there may be an overlap (Gupta *et al.*, 2004). For instance, intellectual stimulation could be an overlap between the two leadership styles. Leaders seek new ways of working, seek opportunities in face of risk and are not likely to support the status quo. Instead of adapting to environmental circumstances, transformational leaders attempt

to create and shape them (Lowe *et al.*, 1996). Finnie and Early (2002) noted that entrepreneurial leaders go beyond the traditional concept of being a charismatic or transformational leader. The need to address the fast-paced and ever-changing business environment gave birth to the concept of entrepreneurial leadership. This ever-evolving dynamic business environment has required leaders to be transformative, as well as innovative and proactive while managing risk (Tarabishy *et al.*, 2005). If a business is facing turbulence, attributed to new venture development, industry changes, or organizational change, entrepreneurial leadership style could lead to the willingness of the leader to take risks and contribute innovative ideas to lead the organization to a point of stability (Cai, 2018). In transformational leadership, the leader arouses follower performance by enticing to the higher follower needs. However, the entrepreneurial leader's ability to evoke such performance is founded in the context of the firm's need to adapt to emerging environmental contingencies. Thus, the basic challenge is to create a willingness in followers to abandon conventional but career-secure activities for creative, entrepreneurial action (Gupta *et al.*, 2004).

It is apparent that entrepreneurial and transformational leadership styles are different (Renko *et al.*, 2015). Transformational leadership is based on inspiring followers to attain organizational goals but also accomplish personal goals. Though individualized consideration is critical to the transformational leadership style, it is not an ingredient of entrepreneurial leadership. Individualized consideration provides support to followers to help them reach self-actualization (Northouse, 2016). Transformational leaders recognize the unique needs and abilities of their employees, treat employees as individuals, build one-to-one relationships and comprehend and are considerate about their differing skills (Avolio and Bass, 1995). Entrepreneurial leaders, consider employees in terms of their entrepreneurial passion and self-efficacy. They enhance followers' beliefs in their own entrepreneurial skills and abilities and ignite the passion for innovation and creativity (Cai, 2018). The center to the entrepreneurial leadership is the emphasis on opportunity-oriented behaviors, by both leaders and those who follow them. Though transformational leadership has some characteristics of such behaviors, they are not endemic.

Leaders with entrepreneurial characteristics may lack inspiration or charisma that is typically characteristic of transformational leaders (Renko *et al.*, 2015). Entrepreneurial leaders also motivate and inspire subordinates through imitation by role modeling instead of inspiring others to personal achievements. Although they lead with clear purpose and goals, entrepreneurial leaders may not be described as charismatic or inspirational by others as often as transformational leaders (Podsakoff *et al.*, 1990). Although a transformational leader uses charisma, inspirational appeals, dramatic presentations, symbolism, or other forms of impression management to inspire admiration, respect, and loyalty, and entrepreneurial leader – charismatic or not – acts as a role model in entrepreneurial behavior, inspiring imitation. In comparing transactional leadership style to the entrepreneurial leadership style, entrepreneurial leadership is more transformational than transactional (Yang, 2008). Some scholars have posited that transformational and transactional leadership styles often depict the leader as someone entrusted with an organizational task that needs to be accomplished within a given organizational context (Gupta *et al.*, 2004). On the contrary, entrepreneurial leadership is founded on the assumption that the leader must enact an entirely emergent organizational task, not one that is carefully planned, and uncover an entirely emergent transaction set, not originally visualized (Gupta *et al.*, 2004; Wang *et al.*, 2012). Thus, under the challenge of high ambiguity and uncertainty, it may have an increasing demand for entrepreneurial leadership (Chen, 2007). These distinct behaviors of the entrepreneurial leadership style making it a distinguished style of leadership whose implications shall be further elaborated.

Knowledge management processes (KMPR)

According to Jain (2007) KM processes are operationally characterized as intentional management procedure to catch, create, store, transform, share and apply both understood and express information to serve workers, organizations and its clients. According to Obeidat *et al.* (2016) in the manifold, rapidly changing and uncertain situation, management of knowledge turns out to be one of the most important and interesting concepts in management. Knowledge management has four main processes including knowledge acquisition, knowledge storage, knowledge sharing and knowledge utilization (Sadeghi and Rad, 2018; Costa and Monteiro, 2016). *Knowledge Acquisition (KA)* refers to the proactive initiatives implemented by the project to identify and collect relevant information from external sources that are critical to its operations (Ali *et al.*, 2018). *Knowledge Storage (KST)* refers to storing, combining, organizing, structuring, getting and connecting advance capacity, for example, reports, phone directories and pictures with knowledge units (Lin, 2013). *Knowledge Sharing (KSH)* refers to acts of making knowledge available to others within the organization (Ipe, 2003, p. 32). *Knowledge Utilization (KU)* is the application of information for the attainment of business goals (Asoh *et al.*, 2007).

Project success (PS)

Project success has been a central topic in project management literature and project success criteria are measures used to evaluate the success or failure of a project (Al-Tmeemy *et al.*, 2011). Operationally, project is defined as a unique and temporary effort that must be completed within a specified budget, time and scope. Jugdev *et al.* (2013) divide project success into two segments, whereby project success elements (critical factors) are related to independent variables that contribute to the project towards the achievement and second segment is project success standards; these are measures used to assess if a project was successful or failure. Shenhar *et al.* (2002) claimed that project success could be influenced by various factors in different types of projects and for that, an organization shall make adjustments in order to deal with the effect of different factors that could affect the chances of project success or failure.

Entrepreneurial leadership and project success

In project-based organizations, effective leadership improves the commitment by instilling a positive attitude and culture that aids in the achievement of project success (Aga *et al.*, 2016). According to Nixon *et al.* (2012), leadership is one of the viable tools utilized by project managers to impact the project result. Under the pressure of competitive situations different project-based organizations, utilize project leaders to react quickly to the environmental changes and achieve the organizational objectives. Existing literature on leadership identifies different leadership styles and their positive effect on project success and management of the organization (Aga *et al.*, 2016; Raziq *et al.*, 2018).

Leaders can benefit from entrepreneurial competencies to deal with the highly turbulent and competitive environment of modern organizations (Luu, 2017). Success through entrepreneurship can be achieved when employees are encouraged and nurtured (Shepherd *et al.*, 2009). Entrepreneurial effort refers to key challenges managers face and is related to the "extent to which top managers are inclined to take business-related risks (the risk-taking dimension), to favor change and innovation to obtain a competitive advantage for their firm (the innovation dimension) and to compete aggressively with other firms (the proactiveness dimension)" (Covin and Slevin, 1988, p. 218).

The relationship between the EL and PS is not clear and has not to be assessed yet, however, literature shows that organizations with a higher level of entrepreneurial

characteristics tend to perform better (Rauch *et al.*, 2009). Thus, understanding project success is as important as understanding EL, and the aspects of both concepts are presented in this section. Project-oriented companies have their results closely related to the success of the projects they develop. In this sense, the evidence from the literature that EL contributes to different performance-related outcomes, for instance, EL style is positively linked with organizational level performance (Abubakar *et al.*, 2018; Al Mamun *et al.*, 2018; Huang *et al.*, 2014; Zainol *et al.*, 2018), the performance of group members towards achieving organizational goals (Renko *et al.*, 2015) and merger and acquisition performance (Strobl *et al.*, 2018). Additionally, like other leadership styles (transformational and transactional leadership styles) that have shown their significant impact on project success (Zaman *et al.*, 2019; Aga, 2016), it can be assumed that EL will have a significant impact on project success. Entrepreneurship has been identified as a well-known path to the achievement of growth and strategic renewal (Hitt *et al.*, 2002). This offers the possibility of a relationship between EL and project success.

In a study developed with leaders and employees, Ahmed *et al.* (2014) showed that entrepreneurial individuals on project teams increase a project's success. Studies also relate project management to innovation (Gordon and Tarafdar, 2007), to the ideas for improvement and development of products, processes, and services (Meredith and Mantel, 2003), to risk-taking (Vezzoni *et al.*, 2013), creativity in solving problems (Al-Tmeemy *et al.*, 2011), to proactiveness (Kerzner, 2004) and change (McLeod *et al.*, 2012) in current ways of doing business. An entrepreneurial leader can provide the team members with an open environment that supports collaboration, knowledge sharing and one where team members can come up with solutions to the problems (Renko *et al.*, 2015) and this can aid during the course of the project and make it more successful (Sohmen, 2013). The aforementioned factors are key characteristics of EL and have been found to have an effect in project-based organizations. Thus, considering the context just presented, the following hypothesis is proposed:

H1. There is a significant impact of entrepreneurial leadership on project success.

Mediating role of knowledge management

Effective leadership can act as a critical instrument in sharing information and knowledge within the organization, so timely and realistic decisions can be taken (Chang *et al.*, 2019). Wei *et al.* (2009) referred to leadership as a KM enabler that enables an organization to align KM behaviors with opportunities, organizational strategies, communicate the best strategies, facilitate the evolution of learning organizations, promote the values of KM, and provide metrics for ascertaining the impact of knowledge. Hence, leadership incorporates strategic issues and how an organization defines its business and uses its knowledge resources to strengthen its competencies (Jager, 1999). Without the commitment of leadership, it is difficult to achieve the success of KM (Civi, 2000). Leaders assume a focal job during the way towards managing organization knowledge. Leaders give vision, motivation, systems and structures at all levels of the organization that improves learning (Bryant, 2003). There are some studies that show the role of leadership in knowledge management processes. For example, Bavik *et al.* (2018) investigated the relationship between ethical leadership and knowledge sharing and found leadership as a key factor influencing the process of knowledge sharing. Singh (2008) assessed the impact of different leadership styles (directive style of leadership, supportive style of leadership, consulting style of leadership and delegating style of leadership) on knowledge management practices in a software firm of India.

KBV recognizes knowledge as the most strategically important of the firm' s resources while EL is committed to the discovery and exploitation of strategic value creation (Gupta *et al.*, 2004). EL is about taking the risk (Kuratko, 2007) while from a knowledge management

perspective, an organization's system that fosters experimentation and risk-taking aids learning, knowledge creation and dissemination (Mikkelsen *et al.*, 2000). Wong and Aspinall (2004) proposed that successful knowledge management requires proactive EL. Their study demonstrated that leadership must create an organizational culture in which there is employee cooperation and knowledge sharing. According to KBV, knowledge is created, stored, and utilized by individuals and not by organizations as a whole, coordinating and integrating the knowledge held by individuals is a difficult task (Miles, 2012). This can be made possible through EL which is characterized by their ability as an integrator and informed (Huang *et al.*, 2014).

KBV identifies knowledge as a key resource and a strategic asset that empowers the business to create value (Zack *et al.*, 2009). In light of KBV (Grant, 1996), successful utilization of knowledge-based resources and effective implementation of KM processes is imperative to achieve better performance (Mazdeh and Hesamamiri, 2014). KBV supports the idea that when knowledge is effectively managed, it creates distinguishing capabilities that contribute to improved performance (Leal-Rodríguez *et al.*, 2013). There is an increasing need for the use of knowledge management as an enabler to improve organization outcomes (Hsiao *et al.*, 2011). With the organizations facing rapid changes both within and outside the organization, knowledge-related capabilities can provide significant support in sustaining competitive advantage and be successful (Torres *et al.*, 2018; Shahzad *et al.*, 2016). For example, KM processes such as acquisition, sharing and utilization not only support learning and decision-making but also drive productivity and profitability through innovative ideas and novelty (Chiu and Chen, 2016).

Entrepreneurial leadership involves a creative and proactive response to environmental opportunities (Brown and Duguid, 1991). From this perspective, entrepreneurial leaders are experts who have learned which problems are relevant to their domain and how to solve them and create a knowledge-based environment that that facilitates problem-solving and value creation (Surie and Ashley, 2008). In a constantly changing external environment, organizational members feel high ambiguity and exert pressure on leaders to respond with new knowledge, skills, and processes (Jansen *et al.*, 2009). From a human capital perspective, entrepreneurial leaders focus on improving an individual's knowledge, skills and capabilities (Leitch *et al.*, 2013); they respond and innovate by broadening existing knowledge and skills (Huang *et al.*, 2014). According to Chen (2007), the entrepreneurial team looks to an entrepreneurial leader for vision, guidance, selection of team members for different assignments, and creating the necessary environment to facilitate innovation, encourage creativity, and facilitate knowledge creation and knowledge sharing. In their role as accelerators, entrepreneurial leaders also make followers rethink their jobs from the perspective of entrepreneurial opportunities that can be recognized through their specific knowledge and skill areas.

Knowledge management processes are rapidly becoming an integral business activity for organizations and it consists of a dynamic and continuous set of processes and practices directed to identifying and leveraging the collective knowledge in an organization to help the organization compete (Garrido-Moreno *et al.*, 2015). Knowledge management plays a significant role in enhancing projects that are used to improve an organization's performance and to sustain a competitive advantage in the market (Jaleel *et al.*, 2019). Therefore, in today's changing environment, management of knowledge is necessary for project success (Polyaninova, 2011). Knowledge management is turning into an imperative vital issue in project management (Yang *et al.*, 2012). KM is critical for project success due to its critical role in knowledge sharing and communication (Solomon, 2001). Through KM, firms look to gain or make conceivably helpful knowledge and to make it accessible to the individuals to impact organizational performance (King, 2009). As per Love *et al.* (2003) poor use of KM adversely influences project success. KM has been displayed as a practice

that helps to improve projects, in this way, KM is seen as the best approach to improve efficiency and increment benefit if knowledge resources are utilized adequately (Alghaila *et al.*, 2017). If organizations desire to maintain good standing in the industry and hold a competitive advantage by a high number of successful project implementations, they should not only rely on the traditional project management improvement techniques but also act to manage their knowledge-of-projects. Moreover, organizations need to employ an efficient KM strategy towards the successful completion of projects. Such a strategy may focus on processes to capture, organize and share the knowledge (Jaleel *et al.*, 2019). KM is important in relation to projects because knowledge related processes affect project performance in terms of costs, schedule, and quality (Berteaux and Javernick-Will, 2015). KM processes (For instance, knowledge sharing and KM utilization) are significant in avoiding the repetition of past mistakes at both the project and organizational levels and improve the chance of project success (Javernick-Will and Hartmann, 2011; Waters and Beruvides, 2012). In addition, organizations face challenges in improving project performance and addressing the risks of project failure, further highlighting the significance of the role of knowledge management and its impact on project success (Anantatmula, 2010). Several studies have identified knowledge management to improve organizational performance (Iqbal *et al.*, 2019; Shahzad *et al.*, 2016). Very few published empirical studies have focused on the direct effect of knowledge management on project performance. While Yang *et al.* (2012) and Yang *et al.* (2017) found a significant impact of KM on project success, the relationship was moderately significant in the study of Fedor *et al.* (2003). Managers in project-based organizations, regardless of their work environment or industry, agree on how important proper knowledge asset management is to project success (Lierni and Ribière, 2008). Researchers have indicated ways in which project knowledge can be protected and utilized to enable individuals to make good decisions using actionable information, experience, and insight (Lierni and Ribière, 2008). Hence, there is a lack of conclusive evidence on the linkage between the two concepts.

Leadership commitment to effective knowledge management in the context of a project-based business strategy is emerging as a potent means of establishing and sustaining competitive advantage (Ajmal *et al.*, 2010) ultimately leading to continued project success. The literature leads to the belief that EL connects with improved KM processes that enhance the chances of project success (see Figure 1). Accordingly, the study argues that KM processes play a mediating role in the relationship between EL and project success. Based on the literature it is proposed that

H2. There is a significant impact of entrepreneurial leadership on KM Processes.

H3. There is a significant impact of KM processes on project success.

H4. KM processes mediate the relationship between entrepreneurial leadership and project success.

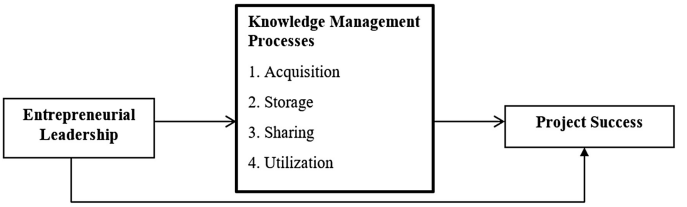


Figure 1.
Research model

Research methodology

Software technology firms make up the sample for the study since these companies probably have a broader spectrum of KM strategies (Donate and Pablo, 2015), In addition, knowledge organizations from technology-intensive industries require a different management approach from non-knowledge organizations, and the role of leadership should be distinctive and supportive of KM processes (Yahya and Goh, 2002). A total of 304 sample size was achieved. The demographic profile of respondents for this study showed that they are typically males, The average age of the respondents was 26.836 years. The majority of the respondents holds a bachelor's degree, work as an android developer and have experienced between 1–10 years. The questionnaire used a seven-point Likert scale from "1" referring to "strongly disagree" to "7" referring to "strongly agree". EL construct compromises eight items ENTRELEAD scale and is adapted from (Renko *et al.*, 2015), example items include *My manager demonstrates passion for my work* and *My manager has a vision of the future of our business*. Knowledge management processes were measured using processes: knowledge acquisition, knowledge storage, knowledge sharing, and knowledge utilization. These dimensions of KMP are measured using 33 adopted items. Knowledge acquisition, knowledge sharing, and knowledge utilization scale were adapted from (Gold *et al.*, 2001). Knowledge acquisition has a total of eight items, sample items include *Our organization has processes for generating new knowledge from existing knowledge* and *Our organization uses feedback from projects to improve subsequent projects*. Knowledge sharing has a total of ten items, example items include *Our organization has processes for transferring organizational knowledge to individuals* and *Our organization has processes for distributing knowledge throughout the organization*. Knowledge utilization has a total of seven items, example items include *Our organization has processes for applying the knowledge learned from errors* and *Our organization has processes for using knowledge to solve new problems*. Knowledge storage is measured using eight items adopted from the study of (Donate and Guadamillas, 2010), example items include *In our organization the organizational procedures are documented through work procedures, written protocols, handbooks, etc.* and *In our organization IT is used to handle, structure, and improve the quality of information and knowledge and encourage communication and accessibility among individuals*. The project success constructs compromises of six items based on the work on Aga *et al.* (2016), example items include *The project was completed on time* and *the project was completed according to the budget allocated*. Data analysis was performed using SmartPLS 3.2.8.

Data analysis and results

Measurement model

The measurement model displays the relationships between the constructs and the indicator variables. As part of the measurement model evaluation, indicators with low factor loadings (< 0.60) were removed (Gefen and Straub, 2005). Only one item (KA1) was removed from the analysis due to low factor loading. The first component of the measurement model is the reliability analysis which includes composite reliability. The desirable cutoff value for the composite reliability is 0.70 (Ringle *et al.*, 2018). Consequently, all the latent constructs of the model possess composite reliability (Table 1). The second component of the measurement model is convergent validity. The measure of convergent validity is the Average Variance Extracted (AVE) for which the cut-off criterion value is 0.50 (Ringle *et al.*, 2018). Hence, constructs possess convergent validity (see Table 1). The component used to access the discriminant validity of the constructs is the Heterotrait Monotrait (HTMT) Ratio procedure. As indicated by Henseler *et al.* (2015), to check for the discriminant validity, the most conservative threshold values of HTMT ratio is less than or

	Λ	Alpha	CR	AVE
Entrepreneurial Leadership		0.895	0.916	0.577
EL1	0.761			
EL2	0.806			
EL3	0.779			
EL4	0.794			
EL5	0.758			
EL6	0.754			
EL7	0.714			
EL8	0.706			
Knowledge Acquisition		0.889	0.914	0.602
KA2	0.813			
KA3	0.797			
KA4	0.798			
KA5	0.805			
KA6	0.705			
KA7	0.730			
KA8	0.777			
Knowledge Sharing		0.922	0.935	0.59
KSH1	0.804			
KSH2	0.825			
KSH3	0.838			
KSH4	0.801			
KSH5	0.747			
KSH6	0.763			
KSH7	0.7500			
KSH8	0.7400			
KSH9	0.699			
KSH10	0.697			
Knowledge Storage		0.914	0.93	0.624
KST1	0.788			
KST2	0.822			
KST3	0.803			
KST4	0.815			
KST5	0.827			
KST6	0.761			
KST7	0.713			
KST8	0.785			
Knowledge Utilization		0.916	0.934	0.669
KU1	0.7700			
KU2	0.898			
KU3	0.869			
KU4	0.843			
KU5	0.842			
KU6	0.797			
KU7	0.6900			
Project Success		0.848	0.888	0.571
PS1	0.703			
PS2	0.775			
PS3	0.803			
PS4	0.765			
PS5	0.799			
PS6	0.678			

Table 1.
Factor loadings, reliability and validity

equal to 0.90. In this study, all the values of HTMT are less than the threshold value of 0.90. Hence, discriminant validity is attained (see Table 2).

Structural model

The structural model displays the relationships (paths) between the constructs on the proposed study model. H1 evaluates whether EL is positively related to PS. The results revealed that EL has a significant impact (total effect) on PS ($\beta = 0.645, t = 13.836, p < 0.0001$). Hence, H1 was supported. H2 evaluates whether EL has a significant impact on KM. The results show that EL has a significant impact on KM ($\beta = 0.777$, $t = 22.554$, $p < 0.001$). Consequently, H2 was supported. H3 evaluates whether KM is positively related to PS. The results revealed that KM has a significant impact on PS ($\beta = 0.624$, $t = 8.343$, $p < 0.001$). Consequently, H3 is accepted. The results are presented in Table 3.

Mediation analysis

Lastly, H4 evaluates whether KM mediates the relationship between EL and PS. The results show that the total effect (H1) was found positive and significant ($\beta = 0.645$, $t = 13.836$, $p < 0.001$). When the mediator was introduced into the model the effect was decreased and the direct relationship was found insignificant ($\beta = 0.160, t = 1.923, p > 0.05$) while the indirect effect with the inclusion of mediator into the analysis was found significant ($\beta = 0.485$, $t = 7.752$, $p < 0.001$). Hence, the results reveal a complete mediation. This shows that the effect of EL on project success passes completely through KM. Consequently, H4 is accepted (see Table 4).

Discussion

The study is set out to establish the relationship between EL, KM processes, and PS. The acceptance of proposed relationships confirms the knowledge-based view. The study found a significant impact of EL on project success. The outcomes affirm this recommendation and

Table 2. Discriminant Validity using HTMT

	EL	KA	KSH	KST	KU	PS
EL						
KA	0.795					
KSH	0.762	0.877				
KST	0.786	0.817	0.89			
KU	0.716	0.722	0.73	0.742		
PS	0.734	0.762	0.752	0.758	0.772	

Table 3. Hypotheses testing

	Path coefficient	Standard Deviation	*T* statistics	*p* values
H1: EL->PS	0.645	0.047	13.836	0.000
H2: EL->KM	0.777	0.034	22.554	0.000
H3: KM->PS	0.624	0.075	8.343	0.000

Table 4. Mediation analysis

	Total effects		Direct effects			Indirect effects		
	Coefficient	*t*-value	Coefficient	*t*-value	Hypotheses	Coefficient	*t*-value	*p* value
EL->PS	0.645	13.836	0.160	1.923	H4:EL->KM->PS	0.485	7.752	0.000

are steady with the discoveries of the past examinations where the positive relationship between EL performance-based outcomes was found (Abubakar *et al.*, 2018; Al Mamun *et al.*, 2018; Renko *et al.*, 2015; Strobl *et al.*, 2018). This shows that a project-based organization characterized by leadership orientation towards risk, challenge, and passion will have a higher propensity for a successful project. This further strengthens the assertion that leaders can benefit from entrepreneurial competencies to deal with the highly turbulent and competitive environment (Gupta *et al.*, 2004). Hence, an entrepreneurial leader can make provision for the team members where they have an open environment to come up with solutions to the problems, support collaboration, and knowledge sharing, (Renko *et al.*, 2015) and this can aid during the course of the project and make it more successful (Sohmen, 2013).

The study found a significant impact of EL on KM processes. The results validated this hypothesis. Wong and Aspinall (2004) suggested that successful knowledge management requires proactive EL. Leadership conduct is an essential factor since leaders enormously affect the direction and effectiveness of KM inside their organizations (Donate and de Pablo, 2015; Singh, 2008). This shows that EL behaves like a dynamic capability, with a focus on the continuous reconfiguration of a firm's knowledge position through initiatives to articulate, codify, and utilize knowledge (Wang and Ahmed, 2007). A strong EL position forces the firm to embark on substantial investment and development initiatives pertinent to knowledge management (Donate and de Pablo, 2015).

The study found a significant impact of knowledge management processes on project success. Results validated this hypothesis and are consistent with the findings of the previous studies (Yang *et al.*, 2012,Yang *et al.*, 2017). The results demonstrate that the project-based organizations that utilize the knowledge management architecture can significantly improve the chances of making a successful project. This certified the validity of KBV that proposes that effective management of knowledge resources can facilitate organizations to achieve superior performance (Grant, 1996). This is achieved since, knowledge management can facilitate communications and provide more informed knowledge to the project manager and project team (Liebowitz and Megbolugbe, 2003).

Finally, the findings of this study provide important empirical insight into the indirect influence of EL on PS through the mediation of KM processes. The results show that KM processes mediate the relationship between EL and project success. The results are consistent with the findings of the previous studies which have found the significant mediating role of KM processes (Donate and de Pablo, 2015; Sadeghi and Rad, 2018; Sarkar *et al.*, 2016). The chain of relationship holds true in light of KBV since the issues with which the KBV concerns itself include the organizational structure, the role of management (Grant, 1996).

Conclusion

The literature demonstrated a generally overlooked, but imperative constructs and ideas, EL and knowledge management processes that guarantee the success of a project. This paper presents one of the earliest attempts to develop and test an integrated model that links EL to project success through the mediating role of KM processes. This research adds to the leadership and KBV literature by exemplifying that EL can significantly enhance the KM processes that have a significant role in improving the chances of a successful project. The findings of this research render that EL is mandatory for successful implementation of KM processes and achieve project success. Policymakers and managers should come up with a deliberate plan to foster EL to encourage KM processes among their employees. Leadership or top management must promote entrepreneurial practices and must provide a clear vision about KM and share it with other members that KM processes are vital for the success of the project.

Implications

Theoretically, the substantiation of the proposed relationships sheds light on the contribution of EL and KM processes in leading to project success. More broadly, the article contributes to the development of studies linking entrepreneurship leadership, knowledge management, and project management, and strengthens the relationship evidenced in the literature that the EL contributes to organizational performance, in this study by contributing to the success of projects, hence, further highlighting the distinctiveness of EL in comparison to other leadership styles. Additionally, the article adds knowledge to studies seeking to understand the factors that influence the success in projects, as well as contributes to the list of benefits of EL for organizations. Taken into consideration the existing gap in the literature that addresses the relationship between EL and project success, this study aids in filling the gap in existing research and stimulate further research in the area. Furthermore, the study ascertains the role EL can play in fostering KM processes in project-based organizations that can use the KM processes to improve the chances of project success. The study explains the pathway through which EL can impact project success. Practically, comprehending the factors that could improve the chances of a successful project, directly and indirectly, could help organizations to get a competitive advantage. The validation of the proposed model points out to managers of the potency of developing EL skillset and improving KM processes in their respective organizations in the pursuit of project success. Thus, the findings suggest that practices of project management can be aligned to the firm's entrepreneurial leadership to enable firms to better result in their projects and generate competitive advantage. Given the proportion of the impact of EL on KM (57.1%) and EL and KM on project success (60.3%), it is important that project managers focus on improving their EL capabilities and improve KM processes within their organization that affect project success.

Research limitations and future research directions

The research study has a certain limitation that must be acknowledged. First, the data has been collected from the employees of the software industry of Pakistan; therefore, the model should be replicated in other contexts. Secondly, in the current study, only one leadership style has been tested, in future studies, other leadership styles should be tested such as authentic leadership, smart leadership, or charismatic leadership. Finally, in future studies with knowledge management processes, knowledge management approaches (social networking, codification, and personalization) should be tested as a mediator between different leadership styles and project success.

References

Abubakar, L.S., Zainol, F.A. and Daud, W.N.B.W. (2018), "Entrepreneurial leadership and performance of small and medium sized enterprises: a structural equation modelling approach", *Journal for International Business and Entrepreneurship Development*, Vol. 11 No. 2, pp. 163-186.

Aga, D.A. (2016), "Transactional leadership and project success: the moderating role of goal clarity", *Procedia Computer Science*, Vol. 100, pp. 517-525.

Aga, D.A., Noorderhaven, N. and Vallejo, B. (2016), "Transformational leadership and project success: the mediating role of team-building", *International Journal of Project Management*, Vol. 34 No. 5, pp. 806-818.

Ahmed, I., Ali, G. and Ramzan, M. (2014), "Leader and organization: the impetus for individuals' entrepreneurial orientation and project success", *Journal of Global Entrepreneurship Research*, Vol. 4 No. 1, pp. 1-11.

Ajmal, M., Helo, P. and Kekäle, T. (2010), "Critical factors for knowledge management in project business", *Journal of Knowledge Management*, Vol. 14 No. 1, pp. 156-168.

Al Mamun, A., Ibrahim, M.D., Yusoff, M.N.H.B. and Fazal, S.A. (2018), "Entrepreneurial leadership, performance, and sustainability of micro-enterprises in Malaysia", *Sustainability*, Vol. 10 No. 5, p. 1591.

Al-Dhaafri, H.S., Al-Swidi, A.K. and Yusoff, R.Z.B. (2016), "The mediating role of total quality management between the entrepreneurial orientation and the organizational performance", *The TQM Journal*, Vol. 28 No. 1, pp. 89-111.

Al-Hakim, L.A.Y. and Hassan, S. (2011), "The role of middle managers in knowledge management implementation for innovation enhancement", *International Journal of Innovation, Management and Technology*, Vol. 2 No. 1, pp. 86-94.

Al-Tmeemy, S.M.H.M., Abdul-Rahman, H. and Harun, Z. (2011), "Future criteria for success of building projects in Malaysia", *International Journal of Project Management*, Vol. 29 No. 3, pp. 337-348.

Alghaila, A.A., Yaob, L., Kiec, C.J. and Alkawsid, J. (2017), "The effect of knowledge management capabilities on project management success", *International Journal of Business*, Vol. 2 No. 2, pp. 1-12.

Ali, I., Musawir, A.U. and Ali, M. (2018), "Impact of knowledge sharing and absorptive capacity on project performance: the moderating role of social processes", *Journal of Knowledge Management*, Vol. 22 No. 2, pp. 453-477.

Anantatmula, V.S. (2010), "Project manager leadership role in improving project performance", *Engineering Management Journal*, Vol. 22 No. 1, pp. 13-22.

Andersen, E.S., Birchall, D., Arne Jessen, S. and Money, A.H. (2006), "Exploring project success", *Baltic Journal of Management*, Vol. 1 No. 2, pp. 127-147.

Asoh, D.A., Belardo, S. and Crnkovic, J. (2007), "Assessing knowledge management: refining and cross-validating the knowledge management index (KMI) using structural equation modeling (SEM) techniques", *International Journal of Knowledge Management*, Vol. 3 No. 2, pp. 1-30.

Bass, B.M. and Avolio, B.J. (1995), *Multifactor Leadership Questionnaire*, Mind Garden, Redwood City, CA.

Bavik, Y.L., Tang, P.M., Shao, R. and Lam, L.W. (2018), "Ethical leadership and employee knowledge sharing: exploring dual-mediation paths", *The Leadership Quarterly*, Vol. 29 No. 2, pp. 322-332.

Berteaux, F. and Javernick-Will, A. (2015), "Adaptation and integration for multinational project-based organizations", *Journal of Management in Engineering*, Vol. 31 No. 6, pp. 1-10.

Brown, J.S. and Duguid, P. (1991), "Organizational learning and communities-of-practice: toward a unified view of working, learning and innovation", *Organization Science*, Vol. 2 No. 1, pp. 40-57.

Bryant, S.E. (2003), "The role of transformational and transactional leadership in creating, sharing and exploiting organizational knowledge", *Journal of Leadership and Organizational Studies*, Vol. 9 No. 4, pp. 32-44.

Butchart, R.L. (1987), *A New UK Definition of the High Technology Industries, Economic Trends*, HMSO, London.

Cai, W., Lysova, E.I., Khapova, S.N. and Bossink, B.A. (2018), "Does entrepreneurial leadership foster creativity among employees and teams? The mediating role of creative efficacy beliefs", *Journal of Business and Psychology*, Vol. 34 No. 2, pp. 203-217.

Chang, Y.-Y., Chao, W.-C., Chang, C.-Y. and Chi, H.-R. (2018), "Transformational leadership influence on unit performance: cross-level moderated mediation evidence", *Leadership and Organization Development Journal*, Vol. 39 No. 4, pp. 554-571.

Chang, Y.-Y., Hodgkinson, I., Hughes, P. and Chang, C.-Y. (2019), "The mediation between participative leadership and employee exploratory innovation: examining intermediate knowledge mechanisms", *Leadership and Organization Development Journal*, Vol. 40 No. 3, pp. 334-355.

Chen, M.H. (2007), "Entrepreneurial leadership and new ventures: creativity in entrepreneurial teams", *Creativity and Innovation Management*, Vol. 16 No. 3, pp. 239-249.

Chen, Y., Tang, G., Jin, J., Xie, Q. and Li, J. (2014), "CEOs' transformational leadership and product innovation performance: the roles of corporate entrepreneurship and technology orientation", *Journal of Product Innovation Management*, Vol. 4 No. 31, pp. 2-17.

Chiu, C.N. and Chen, H.H. (2016), "The study of knowledge management capability and organizational effectiveness in Taiwanese public utility: the mediator role of organizational commitment", *Springer Plus*, Vol. 5 No. 1, pp. 1-34.

Civi, E. (2000), "Knowledge management as a competitive asset: a review", *Marketing Intelligence and Planning*, Vol. 18 No. 4, pp. 166-174.

Costa, V. and Monteiro, S. (2016), "Key knowledge management processes for innovation: a systematic literature review", *VINE Journal of Information and Knowledge Management Systems*, Vol. 46 No. 3, pp. 386-410.

Covin, J.G. and Slevin, D.P. (1988), "The influence of organization structure on the utility of an entrepreneurial top management style", *Journal of management studies*, Vol. 25 No. 3, pp. 217-234.

Cunningham, J.B. and Lischeron, J. (1991), "Defining entrepreneurship", *Journal of Small Business Management*, Vol. 29 No. 1, pp. 45-61.

Dean, H. and Ford, J. (2017), "Discourses of entrepreneurial leadership: exposing myths and exploring new approaches", *International Small Business Journal*, Vol. 35 No. 2, pp. 178-196.

Donate, M.J. and Guadamillas, F. (2010), "The effect of organizational culture on knowledge management practices and innovation", *Knowledge and Process Management*, Vol. 17 No. 2, pp. 82-94.

Donate, M.J. and de Pablo, J.D.S. (2015), "The role of knowledge-oriented leadership in knowledge management practices and innovation", *Journal of Business Research*, Vol. 68 No. 2, pp. 360-370.

Fedor, D.B., Ghosh, S., Caldwell, S.D., Maurer, T.J. and Singhal, V.R. (2003), "The effects of knowledge management on team members' ratings of project success and impact", *Decision Sciences*, Vol. 34 No. 3, pp. 513-539.

Finnie, W. and Early, A. (2002), "Results-based leadership", *Strategy and Leadership*, Vol. 30 No. 1, pp. 23-30.

Freeman, D. and Siegfried, R.L. (2015), "Entrepreneurial leadership in the context of company start-up and growth", *Journal of Leadership Studies*, Vol. 8 No. 4, pp. 35-39.

Fu, L., Liu, Z. and Liao, S. (2018), "Is distributed leadership a driving factor of innovation ambidexterity? An empirical study with mediating and moderating effects", *Leadership and Organization Development Journal*, Vol. 39 No. 3, pp. 388-405.

Garrido-Moreno, A., Lockett, N. and Garcia-Morales, V. (2015), "Exploring the role of knowledge management practices in fostering customer relationship management as a catalyst of marketing innovation", *Baltic Journal of Management*, Vol. 10 No. 4, pp. 393-412.

Gefen, D. and Straub, D. (2005), "A practical guide to factorial validity using PLS-Graph: tutorial and annotated example", *Communications of the Association for Information Systems*, Vol. 16 No. 1, pp. 91-109.

Gil, A.J., Rodrigo-Moya, B. and Morcillo-Bellido, J. (2018), "The effect of leadership in the development of innovation capacity: a learning organization perspective", *Leadership and Organization Development Journal*, Vol. 39 No. 6, pp. 694-711.

Gold, A.H., Malhotra, A. and Segars, A.H. (2001), "Knowledge management: an organizational capabilities perspective", *Journal of Management Information Systems*, Vol. 18 No. 1, pp. 185-214.

Gordon, S.R. and Tarafdar, M. (2007), "How do a company's information technology competences influence its ability to innovate?", *Journal of Enterprise Information Management*, Vol. 20 No. 3, pp. 271-290.

Grant, R.M. (1996), "Toward a knowledge-based theory of the firm", *Strategic Management Journal*, Vol. 17 No. S2, pp. 109-122.

Gupta, V., MacMillan, I.C. and Surie, G. (2004), "Entrepreneurial leadership: developing and measuring a cross-cultural construct", *Journal of Business Venturing*, Vol. 19 No. 2, pp. 241-260.

Harrison, C., Burnard, K. and Paul, S. (2018), "Entrepreneurial leadership in a developing economy: a skill-based analysis", *Journal of Small Business and Enterprise Development*, Vol. 25 No. 3, pp. 521-548.

Henseler, J., Ringle, C.M. and Sarstedt, M. (2015), "A new criterion for assessing discriminant validity in variance-based structural equation modeling", *Journal of the Academy of Marketing Science*, Vol. 43 No. 1, pp. 115-135.

Hitt, M.A., Ireland, R.D., Camp, S.M. and Sexton, D.L. (2002), *Strategic Entrepreneurship: Creating a New Mindset*, Blackwell Publishing, Oxford, UK.

Hsiao, Y.C., Chen, C.J. and Chang, S.C. (2011), "Knowledge management capacity and organizational performance: the social interaction view", *International Journal of Manpower*, Vol. 32 Nos 5/6, pp. 645-660.

Huang, S., Ding, D. and Chen, Z. (2014), "Entrepreneurial leadership and performance in Chinese new ventures: a moderated mediation model of exploratory innovation, exploitative innovation and environmental dynamism", *Creativity and Innovation Management*, Vol. 23 No. 4, pp. 453-471.

Ipe, M. (2003), "Knowledge sharing in organizations: a conceptual framework", *Human Resource Development Review*, Vol. 2 No. 4, pp. 337-359.

Iqbal, A., Latif, F., Marimon, F., Sahibzada, U.F. and Hussain, S. (2019), "From knowledge management to organizational performance: modelling the mediating role of innovation and intellectual capital in higher education", *Journal of Enterprise Information Management*, Vol. 32 No. 1, pp. 36-59.

Jager, M.D. (1999), "The KMAT: benchmarking knowledge management", *Library Management*, Vol. 20 No. 7, pp. 367-372.

Jain, P. (2007), "An empirical study of knowledge management in academic libraries in East and Southern Africa", *Library review*, Vol. 56 No. 5, pp. 377-392.

Jaleel, F., Daim, T. and Giadedi, A. (2019), "Exploring the impact of knowledge management (KM) best practices for project management maturity models on the project management capability of organizations", *International Journal of Management Science and Engineering Management*, Vol. 14 No. 1, pp. 47-52.

Jansen, J.J., Vera, D. and Crossan, M. (2009), "Strategic leadership for exploration and exploitation: the moderating role of environmental dynamism", *The Leadership Quarterly*, Vol. 20, pp. 5-18.

Javernick-Will, A. and Hartmann, T. (2011), "Knowledge management in global environments", in Chinowsky, P.S. and Songer, A.D. (Eds), *Organization Management in Construction* Spon Press, London, UK and New York, NY, pp. 23-40.

Jugdev, K., Perkins, D., Fortune, J., White, D. and Walker, D. (2013), "An exploratory study of project success with tools, software and methods", *International Journal of Managing Projects in Business*, Vol. 6 No. 3, pp. 534-551.

Karmarkar, Y., Chabra, M. and Deshpande, A. (2014), "Entrepreneurial leadership style (s): a taxonomic review", *Annual Research Journal of Symbiosis Centre for Management Studies*, Vol. 2 No. 1, pp. 156-189.

Kantur, D. (2016), "Strategic entrepreneurship: mediating the entrepreneurial orientation-performance link", *Management Decision*, Vol. 54 No. 1, pp. 24-43.

Kerzner, H. (2004), *Project Management: The Best Practices*, 2nd ed., Bookman, São Paulo.

King, W.R. (2009), *Knowledge Management and Organizational Learning Knowledge Management and Organizational Learning*, Springer, Boston, MA, pp. 3-13.

Kuratko, D.F. (2007), "Entrepreneurial leadership in the 21st century: guest editor's perspective", *Journal of Leadership and Organizational Studies*, Vol. 13 No. 4, pp. 1-11.

Leal-Rodríguez, A., Leal-Millán, A., Roldán-Salgueiro, J.L. and Ortega-Gutiérrez, J. (2013), "Knowledge management and the effectiveness of innovation outcomes: the role of cultural barriers", *Electronic Journal of Knowledge Management*, Vol. 11 No. 1, pp. 62-71.

Leitch, C.M. and Volery, T. (2017), "Entrepreneurial leadership: insights and directions", *International Small Business Journal*, Vol. 35 No. 2, pp. 147-156.

Leitch, C.M., Mcmullan, C. and Harrison, R.T. (2013), "The development of entrepreneurial leadership: the role of human, social and institutional capital", *British Journal of Management*, Vol. 24 No. 3, pp. 347-366.

Liebowitz, J. and Megbolugbe, I. (2003), "A set of frameworks to aid the project manager in conceptualizing and implementing knowledge management initiatives", *International Journal of Project Management*, Vol. 21 No. 3, pp. 189-198.

Lierni, P.C. and Ribière, V.M. (2008), "The relationship between improving the management of projects and the use of KM", *Vine*, Vol. 38 No. 1, pp. 133-146.

Lin, H.F. (2013), "The effects of knowledge management capabilities and partnership attributes on the stage-based e-business diffusion", *Internet Research*, Vol. 23 No. 4, pp. 439-464.

Love, P.E., Irani, Z. and Edwards, D.J. (2003), "Learning to reduce rework in projects: analysis of firm's organizational learning and quality practices", *Project Management Journal*, Vol. 34 No. 3, pp. 13-25.

Lowe, K.B., Kroeck, K.G. and Sivasubramaniam, N. (1996), "Effectiveness correlates of transformational and transactional leadership: a meta-analytic review of the MLQ literature", *The Leadership Quarterly*, Vol. 7 No. 3, pp. 385-425.

Luu, T. (2017), "Ambidextrous leadership, entrepreneurial orientation, and operational performance: organizational social capital as a moderator", *Leadership and Organization Development Journal*, Vol. 38 No. 2, pp. 229-253.

McGrath, R.G. and MacMillan, I.C. (2000), *The Entrepreneurial Mindset: Strategies for Continuously Creating Opportunity in an Age of Uncertainty*, Vol. 284, Harvard Business Press, Cambridge, MA.

Masa'deh, R.E., Shannak, R., Maqableh, M. and Tarhini, A. (2017), "The impact of knowledge management on job performance in higher education: the case of the University of Jordan", *Journal of Enterprise Information Management*, Vol. 30 No. 2, pp. 244-262.

Matošková, J., Macurová, L. and Tomancová, L. (2018), "A link between knowledge sharing and managers' characteristics", *Leadership and Organization Development Journal*, Vol. 39 No. 8, pp. 1024-1036.

Mazdeh, M.M. and Hesamamiri, R. (2014), "Knowledge management reliability and its impact on organizational performance: an empirical study", *Program*, Vol. 48 No. 2, pp. 102-126.

McLeod, L., Doolin, B. and MacDonell, S.G. (2012), "A perspective-based understanding of project success", *Project Management Journal*, Vol. 43 No. 5, pp. 68-86.

Meredith, J.R. and Mantel, S.J. (2003), *Project Management a Managerial Approach*, Wiley, New York.

Mikkelsen, A., Ogaard, T. and Lovrich, N. (2000), "Modeling the effects of organizational setting and individual coping style on employees subjective health, job satisfaction, and commitment", *Public Administration Quarterly*, Vol. 24 No. 3, pp. 371-397.

Miles, J.A. (2012), *Management and Organization Theory: A Jossey-Bass Reader*, John Wiley and Sons, San Francisco, CA, Vol. 9.

Mishra, P. and Misra, R.K. (2017), "Entrepreneurial leadership and organizational effectiveness: a comparative study of executives and non-executives", *Procedia computer science*, Vol. 122, pp. 71-78.

Musa, S. and Fontana, A. (2016), "Entrepreneurial leadership measurement validation in innovation management", *ISPIM Innovation Symposium*, The International Society for Professional Innovation Management (ISPIM), p. 1.

Naqshbandi, M.M. and Jasimuddin, S.M. (2018), "Knowledge-oriented leadership and open innovation: role of knowledge management capability in France-based multinationals", *International Business Review*, Vol. 27 No. 3, pp. 701-713.

Nasab, A.H. and Afshari, L. (2019), "Authentic leadership and employee performance: mediating role of organizational commitment", *Leadership and Organization Development Journal*, Vol. 40 No. 5, pp. 548-560.

Nixon, P., Harrington, M. and Parker, D. (2012), "Leadership performance is significant to project success or failure: a critical analysis", *International Journal of Productivity and Performance Management*, Vol. 61 No. 2, pp. 204-216.

Northouse, P.G. (2016), *Leadership: Theory and Practice*, Sage Publications, Los Angeles, CA.

Obeidat, B.Y., Al-Suradi, M.M., Masa'deh, R.E. and Tarhini, A. (2016), "The impact of knowledge management on innovation: an empirical study on Jordanian consultancy firms", *Management Research Review*, Vol. 39 No. 10, pp. 1214-1238.

Podsakoff, P.M., MacKenzie, S.B., Moorman, R.H. and Fetter, R. (1990), "Transformational leader behaviors and their effects on followers' trust in leader, satisfaction, and organizational citizenship behaviors", *The Leadership Quarterly*, Vol. 1 No. 2, pp. 107-142.

Polyaninova, T. (2011), "Knowledge management in a project environment: organisational CT and project influences", *Vine*, Vol. 41 No. 3, pp. 59-69.

Rauch, A., Wiklund, J., Lumpkin, G.T. and Frese, M. (2009), "Entrepreneurial orientation and business performance: an assessment of past research and suggestions for the future", *Entrepreneurship: Theory and Practice*, Vol. 33 No. 3, pp. 761-787.

Raziq, M.M., Borini, F.M., Malik, O.F., Ahmad, M. and Shabaz, M. (2018), "Leadership styles, goal clarity, and project success: evidence from project-based organizations in Pakistan", *Leadership and Organization Development Journal*, Vol. 39 No. 2, pp. 309-323.

Renko, M., El Tarabishy, A., Carsrud, A.L. and Brännback, M. (2015), "Understanding and measuring entrepreneurial leadership style", *Journal of Small Business Management*, Vol. 53 No. 1, pp. 54-74.

Ringle, C.M., Sarstedt, M., Mitchell, R. and Gudergan, S.P. (2018), "Partial least squares structural equation modeling in HRM research", *The International Journal of Human Resource Management*, pp. 1-27, doi: 10.1080/09585192.2017.1416655.

Sadeghi, A. and Rad, F. (2018), "The role of knowledge-oriented leadership in knowledge management and innovation", *Management Science Letters*, Vol. 8 No. 3, pp. 151-160.

Sarkar, R.K., Mostaghimi, M.R. and Akbari, M. (2016), "Investigate the role of knowledge-oriented leadership in innovation and knowledge management practices", *International Business Management*, Vol. 10 No. 11, pp. 2143-2149.

Shahzad, K., Bajwa, S.U., Siddiqi, A.F.I., Ahmid, F. and Raza Sultani, A. (2016), "Integrating knowledge management (KM) strategies and processes to enhance organizational creativity and performance: an empirical investigation", *Journal of Modelling in Management*, Vol. 11 No. 1, pp. 154-179.

Shamim, S., Cang, S. and Yu, H. (2019), "Impact of knowledge oriented leadership on knowledge management behaviour through employee work attitudes", *The International Journal of Human Resource Management*, Vol. 30 No. 16, pp. 2387-2417.

Shenhar, A.J., Tishler, A., Dvir, D., Lipovetsky, S. and Lechler, T. (2002), "Refining the search for project success factors: a multivariate, typological approach", *Randd Management*, Vol. 32 No. 2, pp. 111-126.

Shepherd, D.A., Covin, J.G. and Kuratko, D.F. (2009), "Project failure from corporate entrepreneurship: managing the grief process", *Journal of Business Venturing*, Vol. 24 No. 6, pp. 588-600.

Shujahat, M., Sousa, M.J., Hussain, S., Nawaz, F., Wang, M. and Umer, M. (2019), "Translating the impact of knowledge management processes into knowledge-based innovation: the neglected and mediating role of knowledge-worker productivity", *Journal of Business Research*, Vol. 94, January, pp. 442-450.

Singh, S.K. (2008), "Role of leadership in knowledge management: a study", *Journal of Knowledge Management*, Vol. 12 No. 4, pp. 3-15.

Sohmen, V.S. (2013), "Leadership and teamwork: two sides of the same coin", *Journal of Information Technology and Economic Development*, Vol. 4 No. 2, pp. 1-18.

Solomon, C.M. (2001), "Managing Virtual Teams Technology and people skills create effective teams that might never actually meet", *Workforce-Costa Mesa*, Vol. 80 No. 6, pp. 60-65.

Strobl, A., Bauer, F. and Matzler, K. (2018), "The impact of industry-wide and target market environmental hostility on entrepreneurial leadership in mergers and acquisitions", *Journal of World Business*, doi: 10.1016/j.jwb.2018.03.002.

Surie, G. and Ashley, A. (2008), "Integrating pragmatism and ethics in entrepreneurial leadership for sustainable value creation", *Journal of Business Ethics*, Vol. 81 No. 1, pp. 235-246.

Tarabishy, A., Solomon, G., Fernald, L.W. and Sashkin, M. (2005), "The entrepreneurial leader's impact on the organization's performance in dynamic markets", *Journal of Private Equity*, Vol. 8 No. 4, pp. 20-29.

Torres, A.I., Ferraz, S.S. and Santos-Rodrigues, H. (2018), "The impact of knowledge management factors in organizational sustainable competitive advantage", *Journal of Intellectual Capital*, Vol. 19 No. 2, pp. 453-472.

Vezzoni, G., Pacagnella Junior, A.C., Banzi Junior, A.L. and da Silva, S.L. (2013), "Identification and analysis of critical success factors in projects", *Revista De Gestao E Projetos*, Vol. 4 No. 1, pp. 116-137.

Villaluz, V.C. and Hechanova, M.R.M. (2019), "Ownership and leadership in building an innovation culture", *Leadership and Organization Development Journal*, Vol. 40 No. 2, pp. 138-150.

Wang, C.L. and Ahmed, P.K. (2007), "Dynamic capabilities: a review and research agenda", *International Journal of Management Reviews*, Vol. 9 No. 1, pp. 31-51.

Wang, C.L., Tee, D.D. and Ahmed, P.K. (2012), "Entrepreneurial leadership and context in Chinese firms: a tale of two Chinese private enterprises", *Asia Pacific Business Review*, Vol. 18 No. 4, pp. 505-530.

Waters, N.M. and Beruvides, M.G. (2012), "An empirical study of large-sized companies with knowledge work teams and their impacts on project team performance", *Engineering Management Journal*, Vol. 24 No. 2, pp. 54-62.

Wei, C.C., Choy, C.S. and Yew, W.K. (2009), "Is the Malaysian telecommunication industry ready for knowledge management implementation?", *Journal of Knowledge Management*, Vol. 13 No. 1, pp. 69-87.

Wong, K.Y. and Aspinwall, E. (2004), "Characterizing knowledge management in the small business environment", *Journal of Knowledge Management*, Vol. 8 No. 3, pp. 44-61.

Yahya, S. and Goh, W.K. (2002) "Managing human resources toward achieving knowledge management", *Journal of Knowledge Management*, Vol. 6 No. 5, pp. 457-468.

Yang, C. (2008), "The relationships among leadership styles, entrepreneurial orientation, and business performance", *Managing Global Transitions*, Vol. 6 No. 3, pp. 257-275.

Yang, L.R., Chen, J.H. and Wang, H.W. (2012), "Assessing impacts of information technology on project success through knowledge management practice", *Automation in Construction*, Vol. 22, March, pp. 182-191.

Yang, L.R., Chen, J.H. and Lee, C.H. (2017), "Exploring the links between task-level knowledge management and project success", *Journal of Testing and Evaluation*, Vol. 46 No. 3, pp. 1-18.

Zack, M., McKeen, J. and Singh, S. (2009), "Knowledge management and organizational performance: an exploratory analysis", *Journal of Knowledge Management*, Vol. 13 No. 6, pp. 392-409.

Zainol, F.A., Daud, W.N.W., Shamsu, L., Abubakar, H.S. and Halim, H.A. (2018), "A linkage between entrepreneurial leadership and SMEs performance: an integrated review", *International Journal of Academic Research in Business and Social Sciences*, Vol. 8 No. 4, pp. 104-118.

Zaman, U., Nawaz, S., Tariq, S. and Humayoun, A.A. (2019), "Linking transformational leadership and 'multi-dimensions' of project success: moderating effects of project flexibility and project visibility using PLS-SEM", *International Journal of Managing Projects in Business*, Vol. 13 No. 1, pp. 103-127.

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